Executive summary

Simulation-based decision experiment | Universidad EAN

Bogotá's public health teams face the challenge of prioritizing assessments while balancing missed cases against unnecessary referrals. This academic project explores that decision through a reproducible machine learning experiment using 101,500 published VESPA records. It does not provide a clinically validated screening tool.

The central limitation is explicit: the source file contains no observed alcohol outcome. We therefore generated a simulated binary label from predefined demographic and consumption-setting rules. Positive labels represent simulated alcohol involvement, not confirmed harmful drinking or future risk. Actual substance use may involve multiple substances, so the experimental binary distinction should not be interpreted as a real clinical classification.

Following CRISP-DM, we compared a constant baseline with logistic regression, a decision tree, and a random forest. Preprocessing was fitted inside cross-validation pipelines. Sixty percent of records supported model development, twenty percent supported threshold selection, and twenty percent remained reserved for a single final evaluation. Missing sensitive attributes were retained as unknown rather than probabilistically assigned, and minority sex categories were preserved.

We assumed that a missed positive costs five times as much as an unnecessary positive alert. This ratio is a transparent academic assumption, not an official monetary estimate. The selected model was logistic regression, with a decision threshold of 0.130. On the reserved test set, sensitivity was 99.98%, precision was 64.4%, and weighted error cost was 0.3564 units per record, compared with 0.3567 for the baseline. The alert rate was 99.87%, making workload a central consideration.

The cost advantage over the baseline was small, and its bootstrap interval included zero. Almost universal alerts offer little practical prioritization. The contribution is an auditable workflow demonstrating decision trade-offs. Any measured improvement reflects learning simulation rules; it does not establish public health impact. Before operational use, decision-makers would need observed outcomes, verified analysis units, temporal and person-level validation, agreed costs, service capacity, and fairness review. The recommended next step is data validation and experimental redesign, not deployment.